Mini Project II

"Food Agent AI"

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Introduction

THE PROBLEM

Users waste excessive time manually navigating multiple food delivery apps (Swiggy, Zomato), comparing prices, and repetitively applying filters for location, diet, and budget — a frustrating and inefficient daily task.

- ✗ Multiple apps, no single view
- ✗ Manual price comparison
- ✗ Repetitive filter setup every time

THE SOLUTION

Food Agent AI is an intent-driven autonomous AI agent powered by LLMs and headless browser automation.

1. **Natural Language Input**
   User types: "Find Veg Biryani under ₹300"
2. **AI Understands Intent**
   LLaMA 3.3 parses prompt + user profile
3. **Autonomous Execution**
   TinyFish browses Swiggy & Zomato in parallel
4. **Ranked Results**
   Deals ranked by True Price, shown instantly

Objective

01
Intelligent Web Automation

Develop a context-aware pipeline using the TinyFish SDK for autonomous headless browser control across food delivery platforms.

02
Context Injection via LLM

Parse natural language intents through LLaMA 3.3 70B (Groq API) to generate precise, machine-readable agent commands.

03
True Price Calculation

Rank and compare results after factoring in delivery charges, platform discounts, and applicable taxes for fair comparison.

04
Secure User Profiling

Build a MongoDB + JWT-based system storing user allergies, spice tolerance, delivery address, and budget preferences securely.

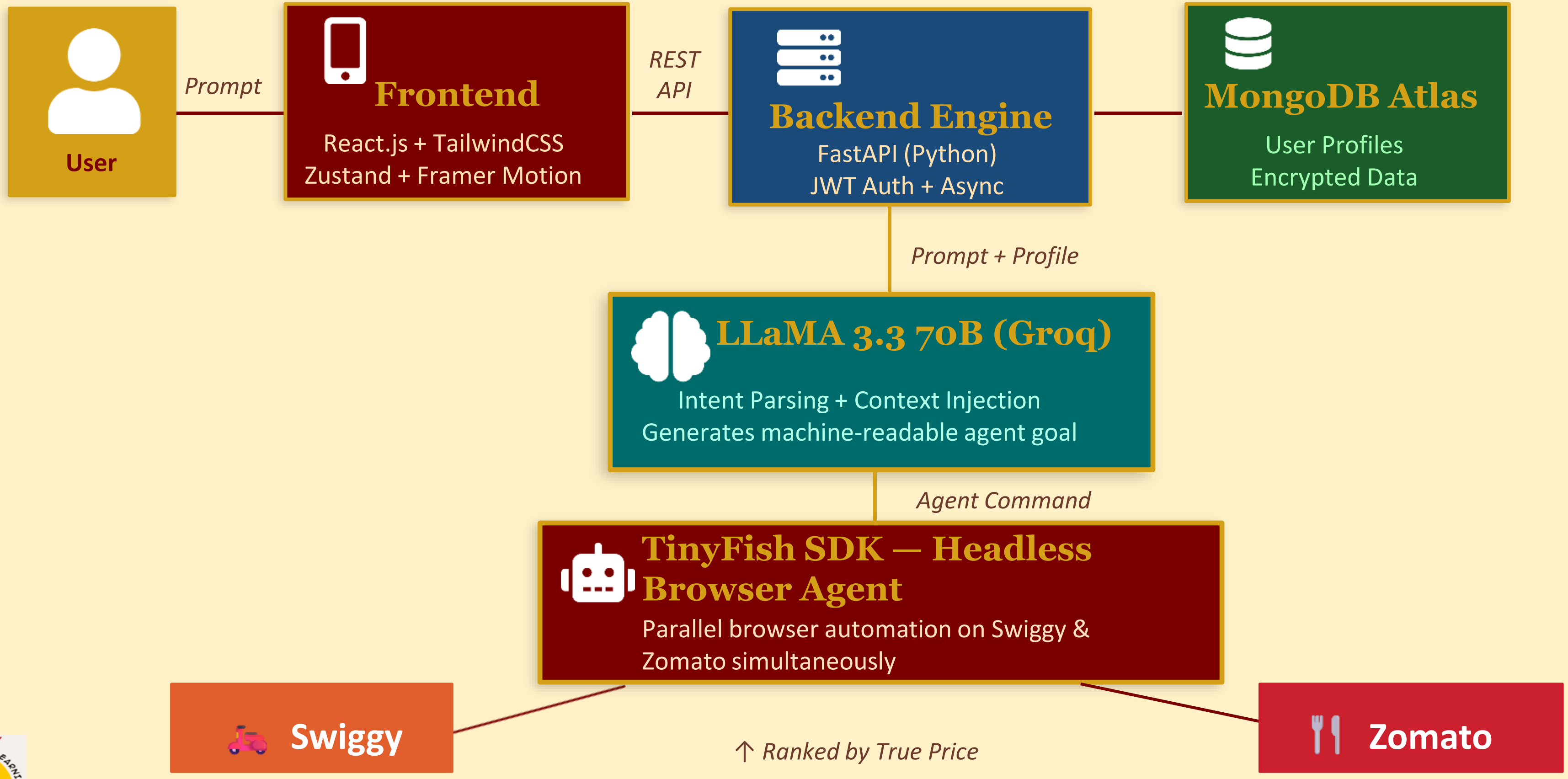
05
Natural Language Understanding

Accept vague, conversational food queries without any structured input, making the system accessible to all users.

06
Zero Manual Filtering

Eliminate repetitive setup of location, diet, and budget filters by automatically injecting user preferences into every request.

System Architecture & Project Flow



Methodology / Project Flow

Frontend

React.js • TailwindCSS • Zustand • Framer Motion

Backend

FastAPI (Python) • JWT Authentication • Async I/O

Database

MongoDB Atlas — Encrypted user profiles & state

AI Engine

LLaMA 3.3 70B via Groq API — Intent parsing

Automation

TinyFish SDK — Headless parallel browser control

Execution Flow

1. **INPUT**
   User submits a vague natural language prompt
2. **INJECTION**
   Backend retrieves user profile (allergies, address) from MongoDB
3. **TRANSLATION**
   LLaMA converts input + profile into strict machine-readable goal
4. **AUTOMATION**
   TinyFish launches headless browsers on Swiggy & Zomato simultaneously
5. **OUTPUT**
   Results extracted, ranked by True Price, displayed to user

Conclusion

LLM as Intent Bridge

Demonstrated that Large Language Models can effectively bridge human intent and complex web UI interactions without manual intervention.

Proactive Safety via Profiles

MongoDB user profiles improved AI safety and speed by injecting allergy avoidance and exact delivery constraints before web execution.

Parallel Execution Success

Achieved simultaneous multi-platform automation — effectively eliminating tedious manual deal comparison across isolated apps.

Future Scope

01 Platform Expansion

One AI agent monitoring Amazon, Flipkart, Meesho ,BookMyShow, Swiggy & Zomato simultaneously — alerting the user the moment a relevant offer appears, faster than any human can browse.

02 Speed Advantage

Detect and act on limited-time instant offers the moment they go live — cart, apply coupon, and stage checkout before any human can even open the app.

03 Automation

Securely log into the user's account via session cookies to auto-stage selected items in the cart — enabling true one-click checkout with zero manual steps.

04 Entertainment

Auto-detect new event listings and book the best available seats the moment tickets drop — beating the manual rush for movies, concerts, and sports events.

05 Transparency

Stream the AI agent's live browser session directly on the frontend — users watch it click, search, and compare in real time, building trust and full transparency.

References

[1] **TinyFish SDK Documentation**
Autonomous Web Agents — browser automation framework for AI-driven tasks.

[2] **Groq Cloud API Documentation**
LLaMA 3.3 70B inference engine — ultra-fast LLM API for production use.

[3] **FastAPI Framework**
Modern, high-performance Python web framework for building APIs.

[4] **MongoDB Atlas**
Cloud-hosted NoSQL database — distributed data platform for applications.

[5] **React.js & TailwindCSS**
Component-based UI development with utility-first CSS framework.

Thank You

Questions & Discussion

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